KODBTW / VIVA PREPARATION

Platform integrations and statistics synchronization

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This guide describes how the current backend fetches coding-platform data, normalizes available metrics and records snapshots. Be precise about which providers are live in the sync path and which remain prototypes.

Contribution reminder: This is a repository-aligned role guide. Describe work you personally performed; confirm individual contribution details with your team.

Explain the system in one flow

Platform account	**THEN**	**Provider adapter**	**THEN**	**Normalize to Platform Stats**	**THEN**	**Sync service**	**THEN**	**Snapshot + sync status**

Core ideas to explain

Adapter boundary

Each provider implements a common adapter contract so the sync service can work with platform-neutral statistics.

Enabled sources

The current normal sync path has live sources for LeetCode, Codeforces and CodeChef.

Provider details

LeetCode uses a public GraphQL query; Codeforces uses its public user.info method; CodeChef uses an unofficial third-party provider.

Persistence

A successful sync saves a source-tagged daily snapshot and updates the platform account’s sync state.

Failure behavior

A failed refresh records failure status and preserves the last successful snapshot rather than replacing it with invented zeros.

Viva quick preparation

Use these as short answer outlines. Explain the idea in your own words and point to the relevant repository code.

Why use an adapter?

It isolates provider-specific requests and parsing behind a shared interface. The application can normalize data without spreading provider details throughout services.

Which integrations are actually enabled?

LeetCode, Codeforces and CodeChef are included in the current live sync path. GeeksforGeeks and HackerRank code in this tree is prototype-level and excluded from normal sync.

What happens if a metric is not provided?

It should remain null or be represented as unavailable, rather than as zero. Zero means a measured value; null means the source did not provide it.

What happens when an upstream call fails?

The sync records a failure state and keeps the previous successful snapshot, so a transient issue does not erase known data.

Can the app guarantee provider availability?

No. Public endpoints and third-party providers can change, rate-limit or become unavailable. CodeChef's provider is unofficial and its permissions/reliability need verification.

Where to point in the repository

```
01 backend/src/main/java/com/kodbtw/adapter/PlatformStatsAdapter.java
02 backend/src/main/java/com/kodbtw/adapter/impl/
03 backend/src/main/java/com/kodbtw/service/PlatformSyncService.java
04 backend/src/main/java/com/kodbtw/dto/PlatformStats.java
05 backend/src/main/java/com/kodbtw/entity/PlatformStatSnapshot.java
06 backend/src/main/java/com/kodbtw/entity/Platform.java
```

30-second presentation

"Platform sync uses an adapter layer to translate each source into a shared statistics model. The current live sync sources are LeetCode, Codeforces and CodeChef, and the sync service stores a source-tagged daily snapshot. If a refresh fails, the system keeps the previous successful snapshot and reports sync status. Some platforms have prototype code only, so I distinguish those from enabled integrations."

Be precise Do not present GeeksforGeeks or HackerRank as live sync integrations. Do not imply ownership, official affiliation, guaranteed uptime or permission for third-party services. Provider data coverage differs.